

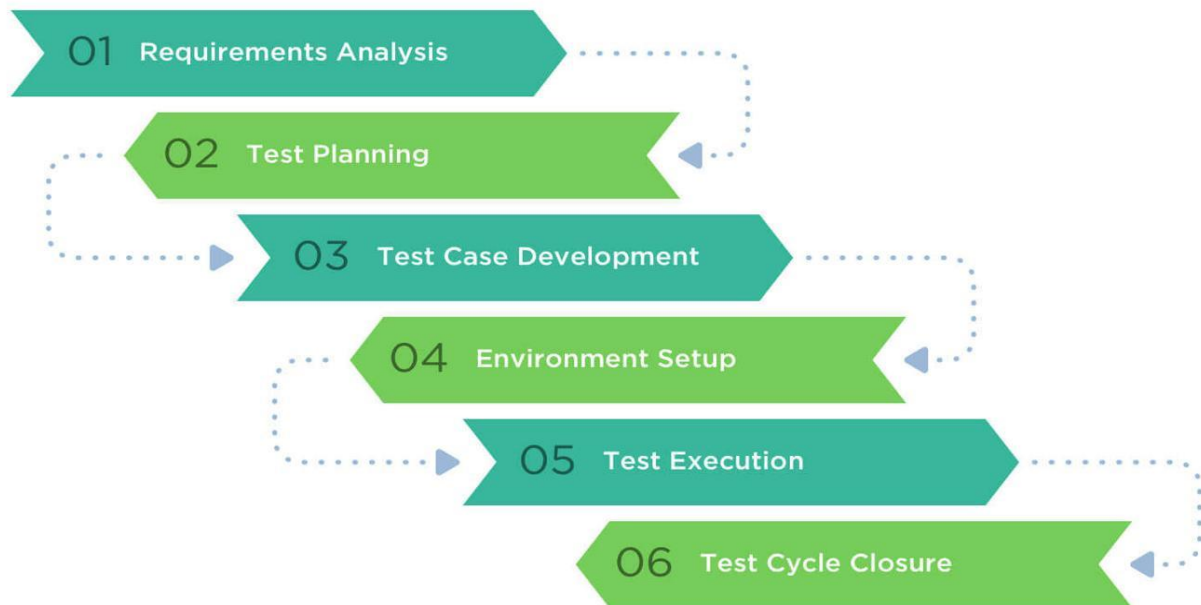
Application Testing Summary

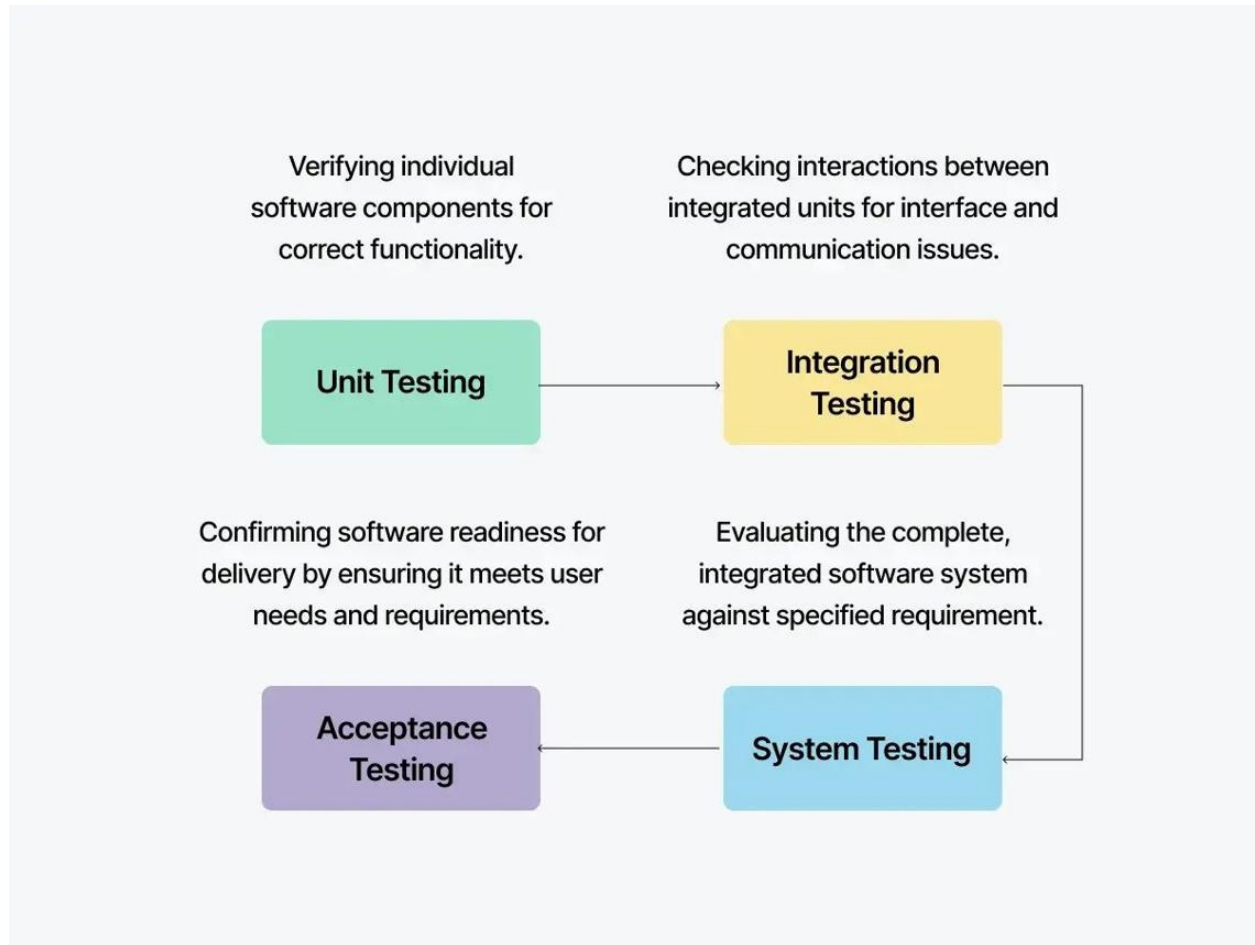
Project: Meal Planning and Recipe Management Web Application

1. Overview

Testing is a critical phase in software development used to ensure that the application functions correctly, meets user requirements, and provides a reliable user experience. For this project, testing was conducted using a structured manual approach focusing on functionality, usability, and performance.

2. Testing Process Flowchart





5

Figure 1: Testing Process Flowchart

Flow Explanation

The testing process follows a structured cycle:

- 1. Identify Test Cases**
Based on system features such as recipe management, meal planning, and authentication
- 2. Execute Tests**
Perform manual testing on each feature
- 3. Verify Results**
Compare expected vs actual outcomes
- 4. Fix Issues**
Resolve bugs found during testing

5. Re-test

Ensure issues are fixed and functionality is stable

3. Testing Approach

3.1 Functional Testing

Functional testing was performed to ensure that all core features operate as expected.

Tested Features:

- Recipe CRUD operations
- Meal plan creation and management
- Nutrition tracker calculations
- Role-based access (admin vs user)

3.2 User Interface Testing

UI testing ensured that the application is visually consistent and responsive across devices.

Checks performed:

- Layout responsiveness (mobile, tablet, desktop)
- Navigation between pages
- Hover effects and animations
- Proper rendering of dynamic content

3.3 API Testing

API endpoints were tested to verify correct data handling and communication.

Validation included:

- Correct JSON responses
- Proper handling of POST, PUT, DELETE requests

- Error responses for invalid inputs

3.4 Authentication and Role Testing

- Admin login tested using demo credentials
- Verified admin-only features (edit/delete) are visible
- Verified standard users cannot access restricted actions

3.5 Nutrition Calculation Testing

- Input values tested with different combinations
- Verified BMR calculations produce accurate outputs
- Confirmed macronutrient values are correctly derived

4. Test Cases and Results

Feature	Test Case	Expected Result	Actual Result
Recipe Creation	Add new recipe	Recipe saved	Success
Recipe Update	Edit recipe	Changes saved	Success
Recipe Delete	Remove recipe	Recipe deleted	Success
Meal Plan	Create plan	Plan stored	Success
Nutrition Tracker	Input data	Correct values calculated	Success
Admin Access	Login as admin	Admin features visible	Success
User Access	Login as user	Restricted access	Success

5. Results

All major functionalities were tested and performed as expected. The system demonstrated stable performance during user interaction, API communication, and data rendering.

Minor UI inconsistencies were identified during testing and resolved to improve overall usability.

6. Limitations

- Testing was primarily manual
- No automated testing frameworks were implemented
- Performance testing under heavy load was not conducted

7. Improvements

- Introduce automated testing (unit and integration tests)
- Implement performance testing tools
- Add end-to-end testing for full workflow validation

8. Summary

The testing process confirmed that the application meets its functional requirements and provides a reliable and responsive user experience. The inclusion of role-based access control and accurate data handling ensures system robustness and usability.